Ghost in the Machine: Neutrino Interactions Without Nuclear Rupture

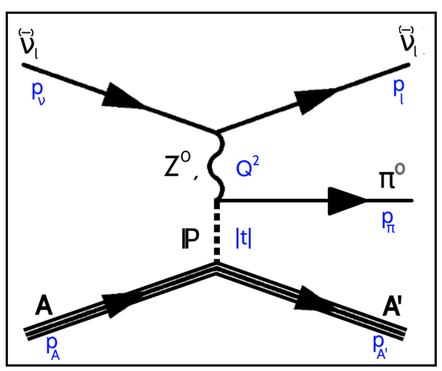
Nimmy Sarah Alex | University of Rochester | MINERvA Collaboration

Neutrinos: The Most Elusive Ghosts

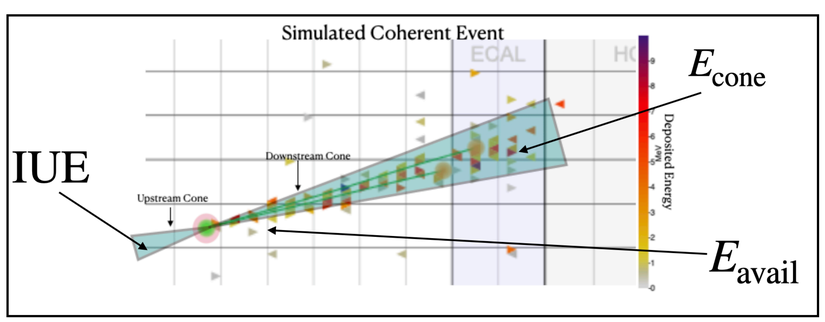
100 billion neutrinos pass through your thumb every second

Nearly impossible to measure.

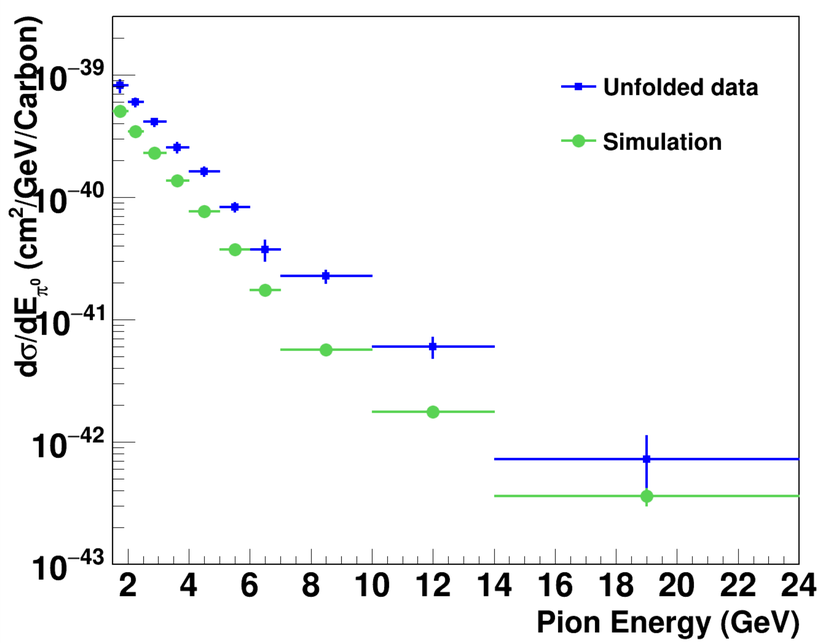
How do we study them?

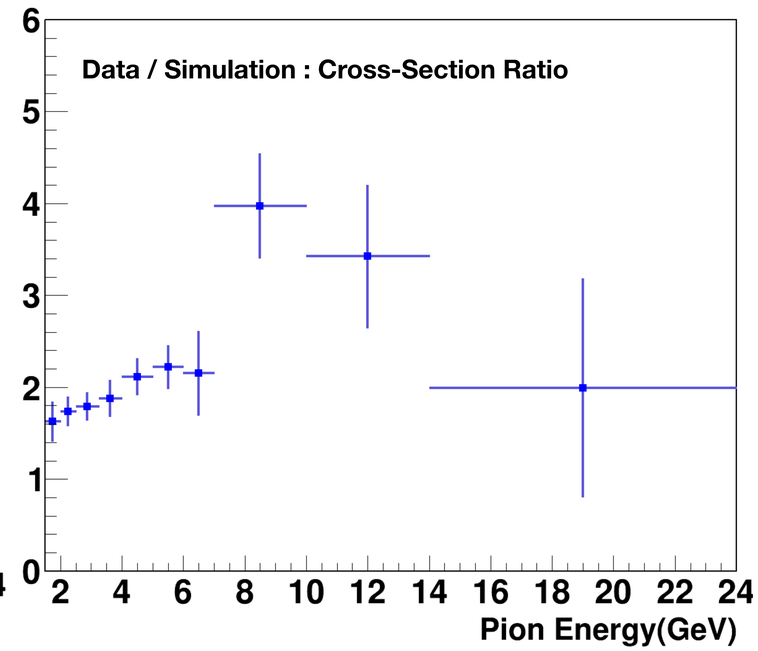


How does this look like in the detector?



Cross-Section : The measure of probability with which a neutrino interacts with a carbon nucleus and produces π^0 is 1.58×10^{-39} cm^2

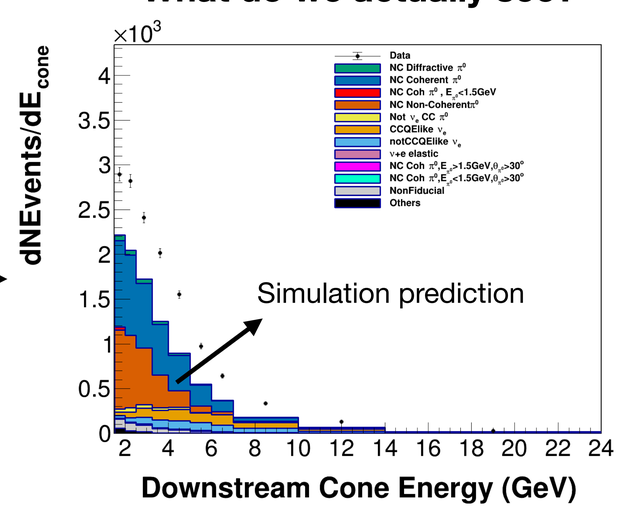




Why does this neutral current $\nu - A$ interaction matter?

- Major background in many neutrino oscillation analysis
- Understand the models used to predict neutrino nucleus interactions
- Constrain uncertainties in future neutrino experiments

What do we actually see?



What does this signify? Analysis Pipeline

Calculate the interaction probability

- Correcting backgrounds
- Translate cone energy to true pion energy
- Correcting detector effects

Further Reading and References